

Health effects of air pollution – evidence and implications

Technical brief



Key messages

- The health effects of air pollution have been documented and described by the medical community for several decades.
- Air pollution can reach almost all organs and systems of the body, impacting respiratory, cardiovascular, cerebrovascular, reproductive and metabolic systems, amongst others.
- Children, pregnant women, older people, people living with chronic disease, and those living in the lowest socioeconomic groups are most at risk.
- Air pollution is the leading environmental health risk factor causing about 6.6 million deaths globally every year.
- Close to 84% of air pollution related deaths are from noncommunicable diseases (NCDs), including heart disease, stroke, chronic obstructive pulmonary disease (COPD) and lung cancer.
- Air pollution burden remains underestimated and is likely to increase, due to population ageing and emerging links with additional diseases and conditions (e.g. low birth weight, neurological and psychiatric diseases, metabolic diseases, cancers other than lung).
- There are substantial inequalities in air pollution levels and health burden globally, closely interlinked with social and economic disparities.
- Ensuring health data are available, reliable and that air pollution exposure is linked with relevant health outcomes using available International Classification of Diseases (ICD) codes for recording, reporting and monitoring diseases is critical for disease surveillance, impact assessment and prioritization of interventions.
- With the enormous health evidence at hand, we must act now to ensure clean air for all and speed up actions for those who are most affected and most vulnerable.

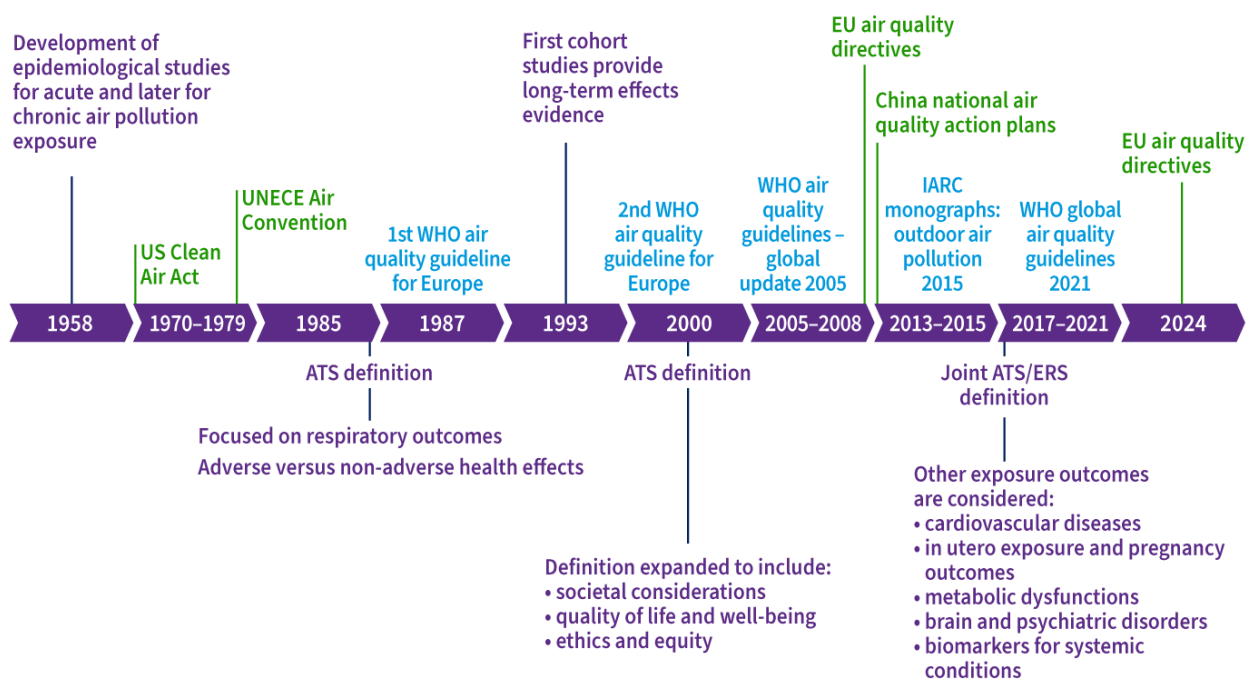


Overview

Health effects of air pollution – a long history

Evidence on air pollution effects on health has been well documented over the last six decades. Important milestones over the years are marked by key epidemiological studies, the first World Health Organization (WHO) reports in the 1950s (1), and the publication of the first edition of the WHO global air quality guidelines in 1987 (2), which provided recommendations for concentrations of air pollutants which should be maintained to protect health. Over the last 40 years, significant statements have been made by medical societies on the definition of what constitutes an adverse health effect of air pollution, which have evolved with mounting scientific evidence from toxicology and epidemiology studies (3, 4) (Fig. 1).

Fig. 1. Timeline of selected air pollution and health milestones



Notes: ATS: American Thoracic Society; ERS: European Respiratory Society; EU: European Union; IARC: International Agency for Research on Cancer.

Air pollution exposure pathways and physiological impacts

While the WHO air quality guidelines series cover a wide range of air pollutants, this summary will focus on the health effects of a subset of pollutants. Particulate matter (PM) is a good indicator of air pollution as it arises from multiple sources, is ubiquitous in our environment and is the pollutant for which there is the largest body of evidence on health impacts. PM consists of a complex mixture of solid particles and liquid droplets of varying size, shape and composition.

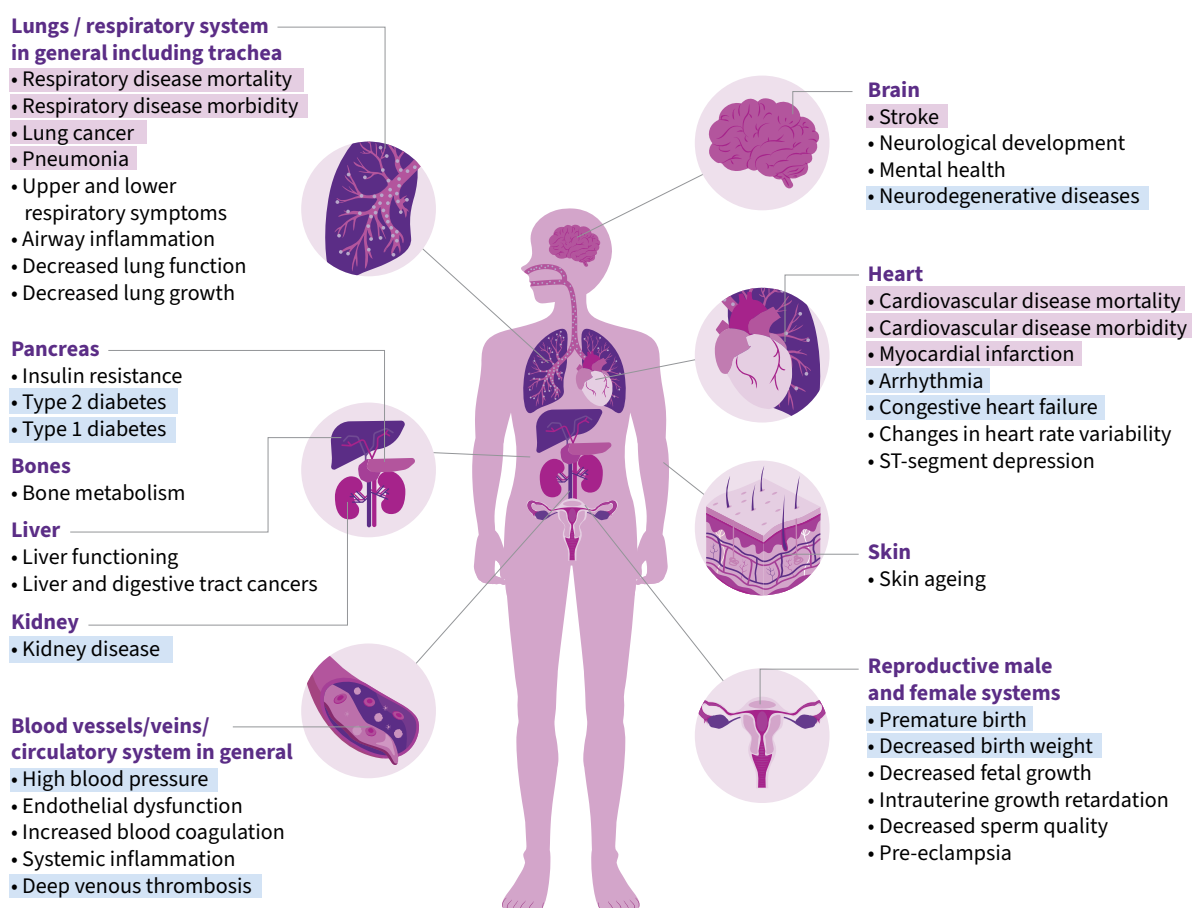
Other main air pollutants are nitrogen dioxide and ground-level ozone, which are gases that originate from various sources and lead to substantial adverse health impacts, though lesser in magnitude with PM (5).

See SPS¹: Exposure to health damaging air pollutants

While inhalation is the main pathway of exposure to air pollutants, there are other routes of human exposure, such as dermal absorption, ocular exposure and ingestion.

When inhaled, health-damaging pollutants including small particles can penetrate deep into the respiratory system and enter the blood stream, allowing them to reach almost all organs within the body (Fig. 2), including but not limited to cardiovascular, reproductive and metabolic systems, brain, liver and kidneys. Many of these damaging effects on the cells and tissues within the human body may not be reversible. Furthermore, air pollution causes oxidative stress, inflammation, immune response, DNA damage, skin damage and direct damage to parts of the body like eyes, nose, throat, lungs, gut, neurons, bones and blood cells.

Fig. 2. Organs and systems affected by air pollution – list of diseases



Source: Adapted from Thurston et al. (3). Disease outcomes highlighted reflect the ATS/ERS assessments of established air pollution adverse effects. It also includes those diseases currently included in WHO assessments (highlighted in pink) as well as noting those for which evidence of an association with air pollution and/or adverse outcomes is emerging (highlighted in blue) (6).

¹ Science and Policy Summary

Short- and long-term exposure to air pollutants and impacts on health

Impact of air pollution episodes on all-cause mortality and more

Short-term, day-to-day exposure to high peaks in air pollution over several minutes (e.g. cycling or walking on very busy streets), hours (e.g. longer commutes in a polluted city) or days (e.g. air pollution episodes in the cities during winter months) presents a significant burden that can cause mild symptoms in healthy people such as runny nose and eyes, cough, headaches or trigger even more severe exacerbations of diseases in chronic lung and heart patients (although it may also occur in persons without a diagnose). This short-term exposure can lead to symptoms that require medication use, doctor visits, hospitalizations, sick days, productivity losses and overall reduced quality of life, or even death.

There are numerous studies investigating effects of daily air pollution levels on all-cause (non-accidental and natural) mortality. Short-term exposure to air pollution is correlated with daily number of deaths. In the latest analyses, including the data from hundreds of cities globally, effects on health remain visible also in less polluted cities (7); despite reductions in air pollution, the effects on health remain high (8). This may be due to multiple factors such as different chemical composition, toxicity and sources of air pollution, as well as socioeconomic determinants or changes in population distribution and susceptibility.

Long-term exposure: the real threat

It is well established that air pollution shortens lives, with life lost ranging from months to years. Longer term exposure over several years, decades or a lifetime increases the risk of premature mortality and can pose greater risks than acute exposure, even at low levels (9). The updated 2021 WHO global air quality guidelines (6) clearly state that the evidence on the association between air pollution and mortality is stronger now, even at lower pollutant levels. This led to the significant lowering of the annual guidelines value to 5 $\mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$ and 10 $\mu\text{g}/\text{m}^3$ for nitrogen dioxide.

Strong associations were also reported in a European multi-country, multi-city study that examined low-level exposure to $\text{PM}_{2.5}$ (8, 10, 11), (approaching 2.5 $\mu\text{g}/\text{m}^3$) and findings suggest that health risks are not linear and there are strong effects at low exposure levels. These findings are in line with other results from around the globe (12, 13). More recently, analyses from the WHO 2024 project Health Risks of Air Pollution in Europe-2 (HRAPIE-2) shows even stronger associations of $\text{PM}_{2.5}$ – but also of PM_{10} – with mortality due to various diseases (14).

In some regions, despite the reduction of air pollution, the population becomes more susceptible as it ages and hence the air pollution attributable burden on mortality can increase. With more high-quality studies, our understanding of the strength and the shape of the exposure-risk relationship is improved, with the latter information being crucial for informing decision-making such as choosing acceptable exposure levels for the population.

Besides leading to premature mortality, long-term exposure to air pollution globally contributes to disease morbidity, including the development of new diseases in millions of people as well as millions of people living with chronic diseases, both of which have debilitating impacts on productivity and quality of life. A recent review confirmed sufficient evidence linking long-term air pollution exposure to various health outcomes, including pediatric and adult asthma incidence, acute lower respiratory infections (ALRI) in children, COPD, and lung cancer, as well as cardiovascular conditions like stroke, hypertension, and ischaemic heart disease. Additionally, emerging—though less certain—evidence suggests associations with autism spectrum disorder in children, diabetes, and dementia (15).

Air pollution: a major risk factor for health

Long-term exposure to air pollution (PM₁₀ and/or PM_{2.5}) is a risk factor for the development of all major NCDs. This section focuses on the morbidity (incidence of disease) outcomes selected by the recent WHO Estimating the morbidity from air pollution and its economic costs (EMAPEC) (16) project, based on a comprehensive review of evidence on long-term exposure to air pollution and morbidity. We refer below to the analyses for the outcomes for long-term effects identified in this project (15).

Respiratory system

The respiratory tract is the first entry point for air pollutants. Exposure to air pollution can adversely affect lungs throughout one's entire lifetime including before birth. Typical health effects of air pollution include a suppressed immune system, inflammation and oxidative stress, impaired lung growth in children, and lung function decrements in children and adults. Long-term exposure to air pollution in healthy people increases the risk of development of a range of respiratory diseases, such as asthma in children (17) and adults (18), COPD (19), ALRIs in children (18), and an increased risk of premature death due to respiratory diseases (9). Furthermore, air pollution presents a substantial burden in the daily life of respiratory disease patients, where exposure to short-term air pollution episodes triggers exacerbations of pre-existing disease, such as cough, shortness of breath, difficulties in breathing, asthma attacks, COPD, increased use of medication, emergency room visits, hospitalizations, and even death.

Cardiovascular system

A wealth of evidence summarized by the American Heart Association in 2010 (4, 20), the European Society of Cardiology in 2015 (21) and the World Heart Federation (22) links air pollution to increased cardiovascular disease morbidity. These statements marked the point when air pollution was recognized as a risk factor for cardiovascular disease. There is now sufficient evidence to conclude that long-term exposure to air pollution is linked to the incidence of ischaemic heart diseases (23), stroke (24) and hypertension (25), as well as premature mortality from cardiovascular diseases (9). Short-term exposure to air pollution also presents a significant burden to patients with heart disease and can trigger severe exacerbations leading to hospitalization and/or death.

Lung cancer

There is clear evidence of a link between long-term exposure to air pollution and incidence (26) and mortality from lung cancer (9). Outdoor air pollution and PM were classified in 2013 as carcinogenic to humans based on evidence of air pollution and lung cancer incidence and mortality by the International Agency on Research for Cancer (27, 28). Other relevant IARC evaluations have also classified diesel engine exhaust (as well as some PM constituents) (29) and household burning of coal as a carcinogen to the human lung (30).

Metabolic effects

A plausible biological mechanism of air pollution promoting diabetes was provided by Sun et al. (31) showing that exposure to particulate air pollution caused increased blood glucose, inflammation in adipose tissue and insulin resistance in high-fat-diet mice. This led to a surge in epidemiological studies on long-term exposure to air pollution and type 2 diabetes, and there is suggestive evidence of a causal relationship between air pollution and type 2 diabetes incidence (32).

Nervous system

Long-term exposure to air pollution can have detrimental effects on the brain via inflammatory reactions and oxidative stress of the brain's nerve cells, which can lead to neurological damage and disease, impairing cognitive development in children and promoting cognitive decline in adults (33). There are ongoing investigations to assess the evidence that air pollution can contribute to autism spectrum disorder in children (34) and dementia incidence in adults (35).

Reproductive male and female systems

Numerous studies have linked air pollution to reproductive functions, pregnancy outcomes, fertility, and fetal health. Beyond suggestive links with birth weight and neonatal deaths (36), air pollution also has been shown to have deleterious effects such as reduced intrauterine growth, premature birth, stillbirth, birth defects in children and pre-eclampsia, gestational diabetes and hypertension in pregnant women. Air pollution can negatively impact human reproduction in both males and females, by increasing time to conceive and decreasing fertility with decreased sperm and egg production.

Emerging evidence

There are ongoing reports on air pollution and a number of mortality and morbidity health outcomes for which evidence is awaiting further epidemiological studies. There is biological evidence showing the direct connection between air pollution exposure and its effects on cells, DNA and the body. Evidence points to linkages between air pollution and COVID-19 morbidity and mortality (37). Air pollution also has been linked to allergic sensitization and increased risk of allergic diseases, including eczema, allergic rhinitis and food allergies.

 **See SPS:** *Climate change, air pollution, pollen and health*

There is now evidence showing that air pollution's adverse effects extend beyond ischaemic heart disease and stroke to increased risk of heart failure and atrial fibrillation. Similarly, research in adults shows possible links with breast, brain, stomach, thyroid, colon, liver, bladder, kidney, leukaemia, lymphoma and pancreatic cancers, as well as leukaemia and brain tumours in children. Other outcomes that are being studied include type 1 diabetes, skin diseases (eczema, skin cancer), eye diseases (conjunctivitis), worsening of rare lung diseases such as cystic fibrosis, chronic kidney diseases, fatty liver diseases, Alzheimer's, Parkinson's disease, multiple sclerosis, and psychiatric outcomes, including depression, anxiety and suicide.

Children's health and other at-risks groups

Although everyone is exposed to air pollution, not everyone is affected equally, and several groups are particularly susceptible and/or vulnerable to its harmful effects. Exposure to air pollution induces adverse health effects, which can vary widely depending on the type, level and duration of exposure, as well as individual factors such as gender, age, vulnerability, education level and socioeconomic status.

 **See SPS:** *Air pollution and children's health*



Key definitions: high-risk groups

Susceptible groups: People with physical predispositions increasing the risks resulting from air pollution exposure. These include age, life stage and pre-existing diseases.

Vulnerable groups: People with increased exposure to air pollution due to external factors such as socioeconomic status, place of residence, occupational exposure, precarious living conditions and polluted energy at home.

Children are more susceptible to air pollution than adults due to several factors: their lungs, brains, and immune systems are still developing, children breathe faster and inhale more air per unit of body weight than adults, and they are more active and spend more time outdoors (38, 39).

Pregnant women are also more susceptible to air pollution due to the special physiological, mental and behavioural factors associated with pregnancy compared with non-pregnant women. Children and adults with chronic diseases, like asthma, cardiovascular and respiratory diseases are also more susceptible to the adverse effects of air pollution, as well as those with weakened immune systems, reduced lung function, impaired cognitive function, weakened circulatory systems and multiple chronic diseases related to ageing. People with the lowest socioeconomic status are also vulnerable to air pollution as they are often more exposed than those with higher socioeconomic status. This can result from various factors, for example, low-income residential areas typically have poorer housing conditions and are often situated in closer proximity to pollution sources increasing their exposure levels. Additional vulnerability factors impacting poorer populations include higher prevalence of other health risks (e.g. higher rates of smoking, obesity and lower rates of physical activity), less access to education and preventive health care and higher likelihood of comorbidities. Outdoor workers, athletes and recreational sports people who spend a lot of time outdoors, especially in polluted environments, are also more vulnerable to air pollution due to the duration and levels of exposure chronically experienced by these populations.

Current exposure levels and health burden

There are large differences in air pollution levels and trends around the globe, and despite the decreasing trends of air pollution exposure in some countries (40), air pollution continues to worsen in many low- and middle-income countries (LMICs).

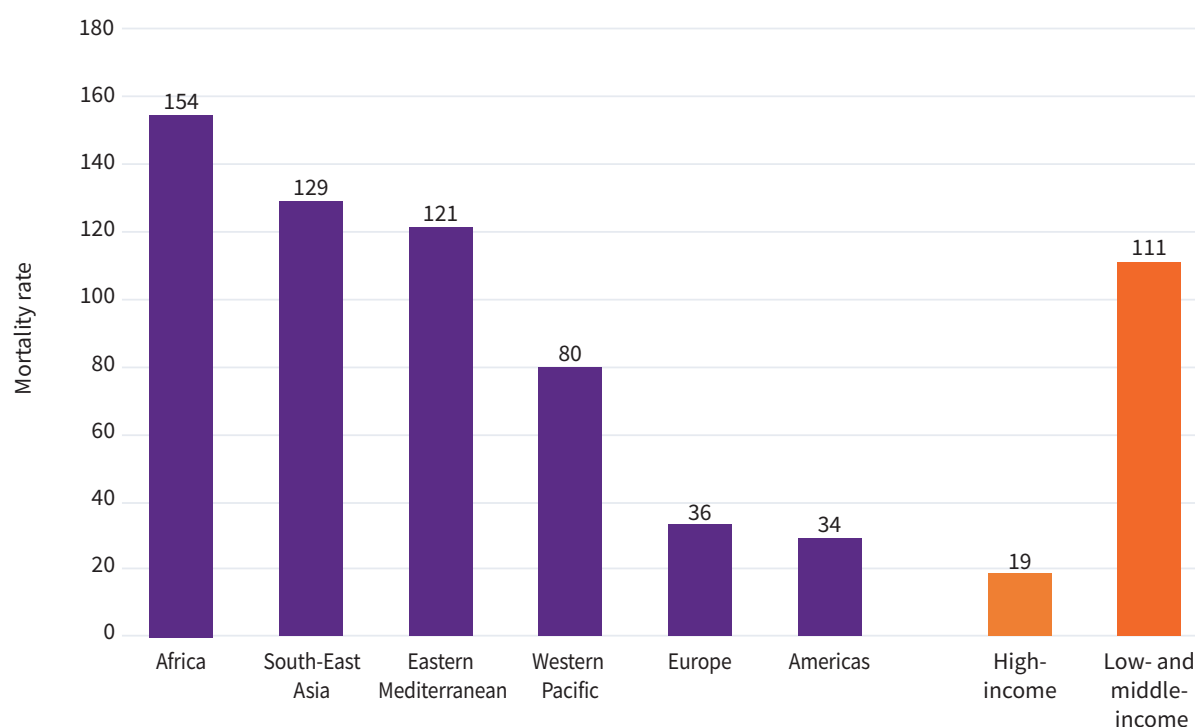
Significant disparities in air pollution levels exist globally, with up to tenfold differences between the cleanest countries in Northern and Western Europe and some of the most polluted ones in Asia (5).

Because of the links with multiple diseases and the ubiquitous nature of the exposure (99% of the world's population is exposed to levels above the WHO guideline for PM_{2.5} of 5 µg/m³), air pollution (all air pollutants combined, including indoor air pollution) is ranked as the second leading risk factor for mortality in 2021, ahead of tobacco smoke and dietary risks (41). Air pollution from PM-attributable deaths – as currently reported under Sustainable Development

Goal Indicator 3.9.1 – accounted for 6.6 million deaths in 2021, including causes causally linked to air pollution: ischaemic heart disease, stroke, COPD, ALRI and lung cancer (42). Additionally, type 2 diabetes and neonatal deaths are considered in the comparative risk assessment of the Global Burden of Disease (41). In total, NCDs account for nearly 84% of the total deaths from air pollution from ambient and household air pollution (42).

The health effects currently considered in global disease burden analyses have a strong evidence base, but those estimates of impacts do not consider health outcomes with emerging evidence such as neurocognitive development in children, neurodegenerative diseases including dementia and Parkinson’s disease, psychiatric disorders, cancers other than lung, chronic kidney diseases, eye and skin diseases, etc. Inhaled air pollution has the capacity to move into the blood and to organs around the body. For example, air pollution particles have been found in cells in the human placenta. Thus, the current global assessments are likely an underestimate of the true air pollution related disease burden, which is also expected to increase, with new studies showing stronger effects of air pollution on health than previously known and at lower levels. It is also striking that the attributable disease burden is unevenly distributed around the world. With an annual rate of around 111 deaths per 100 000 attributed to PM_{2.5}, low- and middle-income countries have a greater disease burden than high-income countries, with a rate of 19 deaths per 100 000. The distribution shown in Fig. 3 indicates a considerably high burden in the WHO African, South-East Asia and Eastern Mediterranean regions (42).

Fig. 3. Global PM_{2.5} attributed age-standardized mortality rates per 100 000 people, by WHO region, 2021



Source: WHO, 2025 (42).

Health risks from low levels of air pollution exposure

Air pollution levels have been decreasing over recent decades in Canada, Europe and the United States, as a result of successful air pollution legislation and/or the implementation of air quality standards (e.g. legal limit values) leading to the introduction of cleaner fuels, regulation of industrial emissions and the implementation of pollution control technologies. This development has generated interest in the health effects of air pollution at low levels, even below legal limit values, in these parts of the world, which has driven additional research. This research has been important to help evaluate whether the current legislation in Canada, Europe, the United Kingdom of Great Britain and Northern Ireland and the United States etc. adequately protects human health, and whether revisions of current legislation are needed, and has influenced recent revisions of air quality standards in the European Union (EU) and United States. On the other hand, in the other parts of the world, such as South-East Asia, North Africa, the Middle East and Latin America, where air pollution levels are much higher and have been stable or increasing, there is, in general, less stringent or even lack of regulation and/or monitoring programmes. These countries also often lack the research infrastructure and capacity for monitoring pollution levels and sources and studying the health effects of air pollution, presenting major obstacles to translating research results into successful policies for air pollution reduction.

Source-specific health effects and composition of PM

PM_{2.5} is a mixture of different chemical components reflecting the contributions of different sources. While there is consensus among scientific studies about the main sources and sectors responsible for PM_{2.5} (43, 44) their contributions vary across studies and locations, which may be due to different assessment methods using dissimilar emission rates, meteorology and physicochemical atmospheric processes. The major global sources of ambient PM_{2.5}, including primary emissions and emission of precursors, are: dust (34%), households (17%), industry (12%), power plants (7%), agriculture (13%), transport (10%) and waste (3%) (45).



See SPS: *Understanding the health impacts of sand and dust storms*



See SPSs: *Sectoral solutions for health topics: Agriculture, Transport, Land use*

The relative importance of these sources differs from region to region. In the African and Eastern Mediterranean regions, dust, followed by household emissions, are the main sources of air pollution. Household energy combustion is also a significant source in the European and South-East Asia regions. In the Western Pacific Region, industry emissions dominate. In the Region of the Americas, natural dust and transport are the main contributors to PM_{2.5} (45–47).

Household PM_{2.5} pollution is mainly associated with residential cooking, heating and lighting using polluting fuels and technologies in LMICs and is highly connected with energy poverty. Other indoor PM sources are emissions from building materials, resuspension and cleaning and consumer products. The contribution of the outdoor environment to indoor PM varies from 10% to 90% and heavily depends on ventilation rates (48).



See SPS: *Household air pollution and related health effects*

Although there is evidence that PM_{2.5} from different sources can have varying toxicities (49), at present there are no clear conclusions to guide regulatory efforts pointing to specific sources of PM_{2.5} as being the specific health effects drivers (50). This implies that the current practice of setting air quality standards based on PM size and concentration still remains relevant. A more effective approach to protecting the public that incorporates source-specific PM_{2.5} needs to be considered as it focuses mitigation on those sources with the greatest adverse health effects. Countries or regions that have successfully reduced air pollution and health effects achieved their targets by controlling the sources that contribute most to air pollution with a focus on cost efficiency but without specific regard to the toxicity of sources or PM_{2.5} components.

Synergies between climate change and air pollution

Climate change and air pollution are intrinsically connected since greenhouse gases and most air pollutants share the same source (i.e. fossil fuel combustion). Additionally, global warming and extreme weather can worsen poor air quality by altering its dispersion, composition and toxicity. Global warming will lead to increases in exposure to ground-level ozone, sand and dust storms, and wildfires, often in combination with extreme heat and droughts, posing additional challenges to health (51). Recent reports of significant synergies in the effects of air pollution (PM_{2.5}, ozone and nitrogen dioxide) and high temperature in exacerbating risk of respiratory and cardiovascular diseases mortality in cities around the globe illustrate the urgent need for clean air actions to mitigate the effects of climate change as well as joint air pollution and climate change actions for optimal health (52, 53).

 **See SPS:** *Health and air pollution co-benefits of climate change mitigation*

Research gaps

Although there is ample evidence on the health effects of air pollution, demanding urgent and ambitious actions around the globe to protect human health, there is additional research needed to address important gaps. Much remains to be learned about how different components of PM_{2.5}, or their sources, are (or are not) linked to various health outcomes. This may include broad categorical components of PM_{2.5}, such as black carbon, ultrafine particles, organic carbon or crustal materials that can be linked to distinctive sources.

There is clear and supportive evidence that exposure to air pollution can lead to increased risk and exacerbation of a range of infectious disease outcomes, and these lead to increases in disease burden. However, because these linkages remain somewhat uncertain, it is important to understand how air pollution exacerbates infectious disease so that its true contribution to the global disease burden is fully accounted for. There is a need for a better understanding of the synergistic effects of air pollution and climate change on health, including the interaction of air pollutants with heat. Finally, targeted policy-oriented research in LMICs on air pollutants and health outcomes would catalyse understanding and build local capacities to enable policy-makers to integrate health evidence into air quality management actions.

Success stories: reducing air pollution improves health

There are several success stories of air pollution reduction as a result of the implementation of regulations/air quality standards or specific measures or policies. Air quality standards in the United States (USA National Ambient Air Quality Standards [NAAQS]) and EU (EU Ambient Air Quality Directive [AAQD]) have led to decreases in air pollution levels in the last few decades. Recent increases in evidence on the health effects of air pollution at low levels (6) have led to revisions and strengthening of US NAAQS and EU AAQD to align them closer to WHO 2021 global air quality guidelines and further protect health. Similar reductions and successful policies are increasingly seen in areas beyond the EU and North America, with China as an example. These illustrate that substantial air pollution reductions can be achieved in a short amount of time with bold and targeted actions and strategies (54).

For example, air pollution levels have been remarkably reduced in China in the last decade since the Air Pollution Prevention and Control Action Plan (2013–2017), Three-Year Action Plan for Blue Skies (2018–2020) and the ongoing Action Plan for Continuous Improvement of Air Quality (2023) were implemented. Air quality monitoring networks have been established nationwide and restrictive control policies implemented at all levels in various government sectors. This has resulted in reductions in anthropogenic emissions in key sectors (e.g. industry, energy and transportation). However, ambient ozone levels remain to be monitored; their slight increase across different regions suggests a future need to understand the sources and source-specific health effects to prioritize monitoring-based pollution control regulatory activities (55, 56).

Still, many areas and countries in the world, most of them with considerable air pollution problems, lack air quality regulation and/or monitoring.

Effective policies and actions reducing air pollution and related health burden

There are a number of examples of policies and actions that can lead to a reduction in air pollution and health benefits. Banning household coal burning in Irish cities in the 1990s contributed to a reduction in respiratory disease mortality (57). Similar benefits can be seen in Krakow, Poland, when authorities banned household coal and wood burning in 2019 (58). The closure of coal power plants in the eastern United States led to significant reductions in air pollution and related mortality (59, 60). Another area of growing interest is policies targeting the reduction of private vehicle traffic and promoting active transportation, such as cycling, walking and using public transport, that are increasingly part of cities' clean air, sustainable transport and climate change action plans. London implemented a congestion charge zone (CCZ) in 2003, and 10 years later introduced an ultra-low emission zone (ULEZ), covering all London boroughs, becoming the world's largest clean air zone (61). The health benefits of LEZs in cities in Germany, Milan, Tokyo and London have been shown for cardiovascular diseases, while CCZs reduced rates of respiratory infections in London (62). The Superblock model is an innovative urban planning strategy implemented in Barcelona (63) – by restricting traffic in several city blocks to streets surrounding the superblock, while within the superblock, car traffic is banned and public space is transformed into green areas, playgrounds, sports facilities, etc. The superblock model makes significant improvements to air quality, reduced noise levels and increased green areas (64).

Reductions in air pollution bring health benefits

It can be difficult to isolate and examine the effects of a specific action on air pollution and health, as improvements in air quality usually result from a combination of legislation and policy measures, technological advancements and individual actions/changes in behaviour. In a systematic review of the reduction in air pollution in 27 LMICs during the COVID-19 pandemic, covering the period January 2019 to March 2022 (65), the authors found that there was a mean percentage change of -30.6 % in $PM_{2.5}$. Despite the majority of global urban growth occurring in LMICs, there are distinct geographical gaps in air pollution data, but where available, health improvements associated with this reduction were reported.

Studies from Switzerland showed that a reduction in PM_{10} helped improve lung function (66) and reduce lung symptoms (67) in adults. A study in the United States investigated the effects of policy-driven air quality improvements on children's respiratory health in southern California (68) and found that a decline in nitrogen dioxide and $PM_{2.5}$ between 1993 and 2014 led to fewer cases of asthma and bronchitis. Recent studies from 2023 and 2024 showed that improvements in air quality in Stockholm, Sweden, between 2002 and 2019, from around 8 to 5 $\mu g/m^3$, contributed to improved lung function (69) and fewer asthma cases in children (70). These Swedish studies have been central to the debate on new EU air pollution limit values, as they show that air quality improvements are effective and contribute to better health, even where air pollution is very low, i.e. around the WHO global air quality guidelines value. Overall, the health benefits of improved air quality remain underexplored.



A girl raises her arms under a clear blue sky, symbolizing health and clean air.
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Way forward



Implement air quality standards and increase sectoral collaboration

Overall, air pollution poses an enormous health burden on people but also huge financial costs to families, health systems and society as a whole. There is ample evidence on the unacceptable health burden of air pollution to support urgent and ambitious actions around the globe for cleaner air, protection of human health and mitigation of climate change. Air quality regulations around the globe should be revised and aligned with the science-based 2021 WHO global air quality guidelines, to provide public health protection. Developing air quality standards that align with the WHO global air quality guidelines or interim targets, as well as air quality monitoring programmes are urgently needed in countries/cities without any. Reduction in air pollution is a complex task requiring a mix of policies and actions across a number of sectors, including transport, energy production, residential heating and cooking, agriculture, industry, waste management and urban planning. The best results are achieved through intersectoral collaboration – involving the health sector – between state governments, regional governments and cities/municipalities, as well as international collaborators.



Strengthen the health sector role in advocacy and data surveillance

For the health sector, there is an urgent need to educate and empower the health workforce to understand and communicate the health risks of air pollution, advise vulnerable groups, engage in multisectoral collaboration (including on health impact assessment of policies) and research, as well as advocate for clean air. Another often overlooked piece of the puzzle is the availability and reliability of health data, which are critical for both programmatic planning, impact assessment, prioritization of interventions and research purposes. Strengthening health data surveillance, civil registration, vital statistics and cause of deaths registration is therefore critical. The use of the International Classification of Diseases (ICD) codes for exposure to air pollution for recording, reporting and monitoring diseases should be further disseminated and used, so as to strengthen documentation of links with major diseases associated with air pollution.



Take sector-specific actions

Within the transport sector, actions include expanding and improving public transport, expanding cycling and pedestrian-friendly infrastructure, limiting car traffic, promoting carpooling and carsharing, electrifying transport and reducing emissions from ships and aircraft/airports.



See SPS: *Transport – sectoral solution for air pollution and health*

Within the energy production and residential energy use sectors, key actions include transitioning away from fossil fuels (coal, oil) towards renewable (wind, solar) energy production, shifting away from polluting fuels and technologies for cooking to clean alternatives, expanding and improving district/central heating, promoting replacements for gas, oil and biomass heating with district heating, solar heating systems and heat pumps, and supporting energy retrofitting to promote energy efficiency and green buildings.

**See SPS:***Phasing out coal-fired electric power generation and implications for public health: Canada: a success story – sectoral solution for air pollution and health***See SPS:***Household energy – sectoral solution for air pollution and health*

Within the agriculture sector it is imperative to reduce air pollutant emissions. Regulations and initiatives should aim to control agricultural waste burning and provide support for sustainable agriculture practices.

**See SPS:***Agriculture – sectoral solution for air pollution and health*

Understanding the complex relationship between air pollution and land use planning is an essential step in developing policies that efficiently reduce the impacts of air pollution on the health and well-being of populations.

**See SPS:***Land use planning: sectoral solutions for air pollution and health*

Urban planning in cities should promote green city development, including parks, street planting, green roofs and walls.

**See SPS:***Green spaces – sectoral solution for air pollution and health*

Waste management should include the implementation of waste sorting, recycling and reuse to reduce the amount of waste dumps (mainly in LMICs) and incineration (high-income countries).

**See SPS:***Open waste burning – sectoral solution for air pollution and health*

Another important step in clean air strategies can be information, education and empowerment of citizens about air pollution levels, sources and health effects. This will enable them to reduce exposures, minimize their own risk, reduce emissions and empower themselves to demand clean air actions from policy-makers. This can be accomplished by early education of children and youth in schools, training of health professionals on how to assess air pollution in clinical practice, and educating patients to empower them to avoid risk.

There are several individual measures that can be taken to reduce exposure to air pollution, such as choosing less polluted streets during commutes, especially

when walking and cycling; ventilating homes with windows facing the least polluted streets/backyards; avoiding using wood/coal burning for heating, lighting and cooking. The use of face masks outdoors and air purifiers can help reduce exposure to air pollution, in very specific situations. However, there is generally poor evidence on the effectiveness of these measures. Yet, in the hierarchy of interventions, the most efficient, sustainable and equitable measure to protect health from air pollution remains to reduce emissions at the source.



Support international agreements

Countries should support and pursue ambitious goals in significant international and regional agreements, such as the Gothenburg Protocol under the Convention on Long-Range Transboundary Air Pollution, which sets reduction commitments for various pollutants, and under the International Maritime Organization, which sets emission standards for shipping.

Methodology for content development

WHO defined the scope of the document and collaborated with its Advisory Groups (Scientific Advisory Group on Air Pollution and Health [SAG]; and the Global Air Pollution and Health – Technical Advisory Group [GAPH-TAG]) members, which cover a wide range of expertise, to prepare the initial draft based on an overview of health effects of air pollution. Evidence for this rapid review was gathered and presented at monthly discussions with experts involved on the development of the document.

WHO staff and consultants from the WHO Air Quality, Energy and Health Unit reviewed the report to ensure alignment with the WHO requirements for the collections of WHO Air Quality, Energy and Health Science and Policy Summaries. This series synthesizes current knowledge and evidence on air quality, energy access, climate change links and health, primarily to inform intergovernmental discussions.

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Declarations of interest

All external experts submitted a declaration of interest to WHO disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the report. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects covered by the report.

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